Arista Access Node KOR-ACC-79 Hardware/Environmental Fault Failure Resolution SOP

Node: KOR-ACC-79 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8787

Description

The KOR-ACC-79 access node in the Koramangala Datacenter is an Arista-equipped device operating within the realm of an access domain, providing connectivity to edge devices and customers. The Arista CLI dialect is employed for all diagnostic and resolution tasks. Hardware/environmental faults such as ASIC memory parity errors and line card crashes can occur in this role due to the inherent complexities of modern networking systems, including the intricacies of ASIC memory and the potential for thermals to impact line card operation. Furthermore, the loss of power supply redundancy in the access node can be a precursor to such failures, as it increases the likelihood of single-point-of-failure scenarios. The ALARM_8787 alarm code, characterized as a major severity event, necessitates immediate attention to prevent service disruptions and data loss.

Pre-requisite Checks

- Verify remote hands have been dispatched to the Koramangala Datacenter
- Confirm the status of redundant links connecting KOR-ACC-79 to adjacent nodes
- Check power supply redundancy for KOR-ACC-79 and ensure proper functioning

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash due to thermal issues	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal memory error, Expected Output: memory error details, including ASIC memory parity errors, Description: This command retrieves information about memory errors occurring within the system, allowing the identification of potential ASIC memory parity errors.

Command: show hardware linecard detail, Expected Output: line card details, including temperature readings and fan status, Description: This command provides a detailed view of the line card's status, enabling the assessment of potential thermal issues or fan malfunction.
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```
Command: show power redundancy status, Expected Output: power redundancy status, including current power supply status and redundancy levels, Description: This command displays the current power redundancy status, facilitating the identification of any issues with power supply redundancy.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC memory, Command: replace system internal memory, Description: Replace the faulty ASIC memory to resolve the parity error and ensure system stability.

Step 2: Step: Replace line card with thermal issues, Command: replace hardware linecard, Description: Replace the line card experiencing thermal issues to prevent further damage and ensure smooth operation.

Step 3: Step: Add power supply redundancy, Command: configure power redundancy add, Description: Add an additional power supply to ensure power redundancy and prevent single-point-of-failure scenarios.

Post-Check Verification

```
Command: show system internal memory error, Expected Output: no memory error details, Description: Verify that the memory error issue has been resolved.

Command: show hardware linecard detail, Expected Output: no line card issues, Description: Verify that the line card has been replaced successfully and is functioning correctly.

Command: show power redundancy status, Expected Output: power redundancy status indicating multiple power supplies are online, Description: Verify that power redundancy has been added and is functioning correctly.
```

Roll-back Procedures

- Step: Revert to previous power supply redundancy configuration, Command: configure power redundancy revert, Description: Revert to the previous power supply redundancy configuration to undo the change.

- Step: Replace original line card, Command: replace hardware linecard original, Description: Replace the original line card to undo the replacement.

- Step: Reinstall original ASIC memory, Command: replace system internal memory original, Description: Reinstall the original ASIC memory to undo the replacement.